

	1	2	3	4
	DESIGNATION			
A	COVER SHEET	1	X	
	SUMMARY	2	X	
	SYMBOLS	3	X	
	LEGEND	4	X	
B				
	POLARITY DISTRIBUTION	7	X	X
	INCOMING CHARGER	8	X	X
	HAND CIRCUIT BREAKER	9	X	X
	INCOMING BATTERY	10	X	
C	HAND CIRCUIT BREAKER	11	X	X
	BUS BAR COUPLING	12	X	
	HAND SWITCH	13	X	
	OUTGOING	14	X	
	HAND CIRCUIT BREAKER	15	X	X
D				
	SWITCH POSITION AND FUSE BLOWN	16	X	X
	GROUPED FAULT	17	X	X
E				
	METERING AND PROTECTIVE SYSTEM	18	X	X
	HEATERS AND LIGHTING CUBICLES	19	X	X
F				

COMMENTS :

REVISION

09

AS BUILT

N.	Date	N.	Date	N.	Date	AS BUILT	ASB
D	95/08/09	HAR	95/08/09	DEG	95/08/09	AS BUILT	ASB
C	94/01/26	HAR	94/01/26	DEG	94/01/26	UP DATED ACCORDING TO MANUFACTURING & GOOD FOR EXECUTION	GFE
B	93/02/26	HAR	93/02/26	DEG	93/02/26	UP DATED ACCORDING TO EM 303 AND MANUFACTURER DATAS	GFE
A	92/12/3	HAR	92/12/3	DEG	92/12/3	ORIGINAL ISSUE	APP
REN.	ETABL. ESTABLISHED	VERIFIE CHECKED	APPROUVE APPROVED			MODIFICATIONS	STAT.

VINH-SON HYDRO-POWER STATION

2 x 40 MVA

CONTRACT N° 12-05/CEG-PC3-PET/01-91

SOCIALIST REPUBLIC OF VIET-NAM
MINISTRY OF ENERGY
PETECHIM-POWER COMPANY Nr.3
ENGINEER EDF

LEADER: **CEGELEC** FRANCE

ALSTHOM - JEUMONT
CEGELEC

NEYPIC
SDEM

ECHELLE
SCALE

POWER PLANT

48V LOW VOLTAGE DC SYSTEM

TYPICAL ELECTRICAL DIAGRAM

	DATE	NOM NAME	VISA
ETABL. ESTABLISHED	92/12/3	XTO	
VERIFIE CHECKED	92/12/3	HAR	
APPROUVE APPROVED	92/12/3	DEG	

CEGELEC

Division ERE P : 35, rue d'Alsace F - 92531 Levallois Perret Cedex

CGE VIN00 SLCA-ST001

P2135A30113-D

FOLIO FIN
SHEET 01 19

1		2		3		4						
DESIGNATION		F9	REVISION									
			A	B	C	D						
A	COVER SHEET	1	X	X	X	X						
	SUMMARY	2	X	X	X	X						
	SYMBOLS	3	X									
	LEGEND	4	X									
B												
	POLARITY DISTRIBUTION	7	X	X								
	INCOMING CHARGER	8	X	X	X							
	MANUAL CIRCUIT BREAKER	9	X	X	X							
	INCOMING BATTERY	10	X									
	MANUAL CIRCUIT BREAKER	11	X	X	X							
C	BUS BAR COUPLING											
	MANUAL SWITCH											
	OUTGOING											
	MANUAL CIRCUIT BREAKER											
	SWITCH POSITION AND FUSE BLOWN	14	X	X								
D	GROUPED FAULT	15	X	X								
	METERING AND PROTECTIVE SYSTEM	18	X	X	X							
E	HEATERS AND LIGHTING CUBICLES	19	X	X								
F												

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5		6		7		8						
DESIGNATION		F9	REVISION									
			A	B	C							
SWITCH	DISCONNECTOR	1										
	CIRCUIT BREAKER	2										
	CONTACTOR	3										
	CONTACTOR	4										
	CONTACTOR	5										
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	CONTACTOR	100										

SUMMARY

CEGELEC
P2135A30113-D
02 03

BUIL21ET303

ANSI	DESIGNATION	CEI	ANSI	DESIGNATION	CEI
12	OVERSPEED DEVICE	n >	50-76	OVERCURRENT RELAY	I >
13	SYNCHRONOUS-SPEED DEVICE	u >	50N	OVERCURRENT RELAY, NEUTRAL	IN >
14	UNDERSPEED DEVICE	c >	51	TIME-LAG OVERCURRENT RELAY	I >
15	SPEED MATCHING DEVICE	c >	51	INVERSE TIME OVERCURRENT RELAY	I >
21	DISTANCE OR UNDERIMPEDANCE RELAY	z >	51N	TIME-LAG OVERCURRENT RELAY, NEUTRAL	IN >
21	DISTANCE OR UNDERREACTANCE RELAY	x >	51V	TIME-LAG OVERCURRENT RELAY WITH VOLTAGE RESTRAINT	I > U >
21	DISTANCE OR OVERADMITTANCE RELAY	y >	53	EXCITATION INCREASE RELAY	B >
23	TEMPERATURE CONTROL DEVICE	θ	55	POWER FACTOR RELAY	cos φ
25	VOLTAGE COMPARISON DEVICE	u >	56	EXCITATION RELAY	B
25	PHASE-ANGLE COMPARISON DEVICE	α >	58	RECTIFICATION FAILURE RELAY	Δ > 0
25	FREQUENCY COMPARISON DEVICE	f >	59	OVERVOLTAGE RELAY	U >
25	COUPLING DEVICE	↑	59N	NEUTRAL OVERVOLTAGE RELAY	UN >
25	SYNCHRONIZING OR SYNCHRONISM-CHECK DEVICE	UF	59-81	OVERINDUCTION RELAY	U(f)
26	THERMOMETER DETECTOR	θ	60	VOLTAGE BALANCE RELAY	Ud
27	UNDERVOLTAGE RELAY	u >	60	CURRENT BALANCE	Id
27	NO-VOLTAGE RELAY	u = 0	63	PRESSURE SWITCH	P
30	ANNUNCIATOR RELAY	⊗	64	FRAME LEAKAGE RELAY	h
32	DIRECTIONAL ACTIVE POWER RELAY	Δ >	67	DIRECTIONAL OVERCURRENT RELAY	I >
32	DIRECTIONAL REACTIVE POWER RELAY	Δ >	67N	DIRECTIONAL HOMOPOLAR OVERCURRENT RELAY	IN >
32	REVERSE ACTIVE POWER RELAY	Δ >	71	LEVEL SWITCH	○
32	REVERSE REACTIVE POWER RELAY	Δ >	74	ALARM RELAY	e
37	UNDERCURRENT RELAY	<	78	MEASURING RELAY AND PHASE-ANGLE PROTECTION	→
37	UNDER POWER RELAY	P <	79	RECLOSING RELAY	f
39	MECHANICAL CONDITION MONITOR	Δ >	81	FREQUENCY RELAY	f
40	EXCITATION FAULT RELAY	B >	87	DIFFERENTIAL PROTECTIVE RELAY	Δ I
40	NO-EXCITATION RELAY	B = 0	87N	DIFFERENTIAL PROTECTIVE RELAY	Δ I
46	PHASE-BALANCE OR REVERSE-PHASE CURRENT RELAY	I >	95	BUCHHOLZ RELAY	Δ I
47	VOLTAGE RELAY	u >			
49	THERMAL RELAY	θ			
49	THERMOSTAT	θ			

SWITCH	RESIDUAL CURRENT TRANSFORMER	CAPACITOR	MANUALLY OPERATED
DISCONNECTOR	SHUNT	CAPACITOR VOLTAGE DIVIDER	ELECTROMAGNETIC ACTUATOR
CIRCUIT BREAKER	INDICATING INSTRUMENT	INDUCTOR WINDING	PNEUMATIC OR HYDRAULIC CONTROL
CONTACTOR	RECORDING INSTRUMENT	BATTERY	REVERSER
CONTACTOR POLE (contact closed in the unoperated position)	ENERGY METER	CONVERTER OR MEASURING TRANSDUC.	KEY ALWAYS FREE
SWITCH DISCONNECTOR POLE (on load isolating switch)	IMPULSE COUNTER	RECTIFIER	KEY FREE FOR THE POSITION IN THE DIAGRAM
FUSE	MEASURING RELAY OR RELATED DEVICE	PLUG AND SOCKET	KEY IMPRISONED FOR THE POSITION IN THE DIAGRAM
THERMAL EFFECT MAGNETIC EFFECT	SIGNAL LAMP	ELECTRICAL CONTROL	ISOLATED PHASE BUS DUCT
THREE WAYS SWITCH	LIMITER	LINK	CABLE
ROTATING MACHINE	DOUBLE DISCONNECTOR	INVERTER	OVERHEAD LINE
TRANSFORMER WITH 2 WINDINGS	SPARKGAP	EARTH MASS	CARRIER LINE TRAP
TRANSFORMER WITH 3 WINDINGS	LIGHTNING ARRESTER	PULSE	PLUG (male)
AUTO-TRANSFORMER	RESISTOR	MECHANICAL INTERLOCK	SOCKET (female)
CURRENT TRANSFORMER	VARIABLE RESISTOR	KEY OPERATED	POWER LINE CARRIER COUPLING DEVICE

SYMBOLS

P2135A30113- A

Division
ERE P

03 04

BULZLEI 309

LEGEND AND ELECTRICAL COMPONENTS

BP PUSH BUTTON
ON:GREEN STOP:RED RESET FAULT:YELLOW
TEST LAMP:BLEU OTHER:WHITE

LA LAMPS
ON:RED STOP:GREEN FAULT:YELLOW OTHER:WHITE

FU FUSES

XR AUXILIARY RELAY

CD FAULT AUXILIARY CONTACT

CA POSITION AUXILIARY CONTACT

FCD DISCONNECTED POSITION CONTACT

FCE CONNECTED POSITION CONTACT

FC LIMIT SWITCH

JD CIRCUIT BREAKER

JC CONTACTOR

JS DISCONNECTING SWITCH

EA TRIPPING COIL

YI CURRENT MEASUREMENT RELAY

YU VOLTAGE MEASUREMENT RELAY

YT TEMPERATURE MEASUREMENT RELAY

YG ISOLATION MEASUREMENT RELAY

MU VOLTAGE TRANSDUCER

MI CURRENT TRANSDUCER

MW POWER TRANSDUCER

II CURRENT INDICATOR

IU VOLTAGE INDICATOR

BM TEST BOX

TU VOLTAGE TRANSFORMER

TI CURRENT TRANSFORMER

CC CONTROL SWITCH

CONTROL

SIGNALING

TEST LAMP

0010P

POLARITE LEGEND TRIBUTION

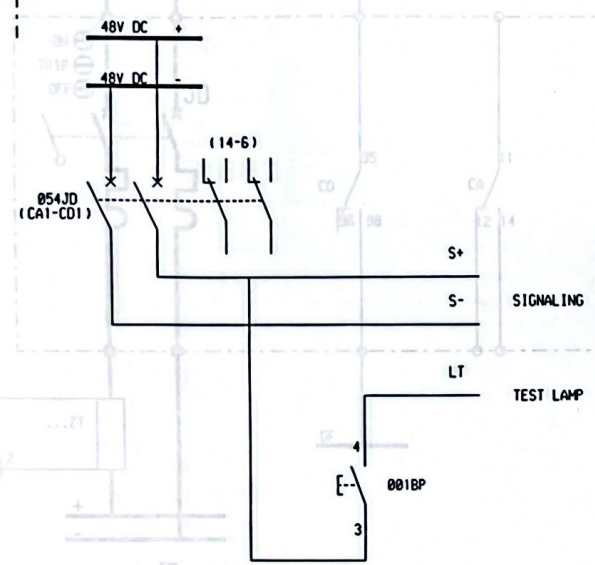
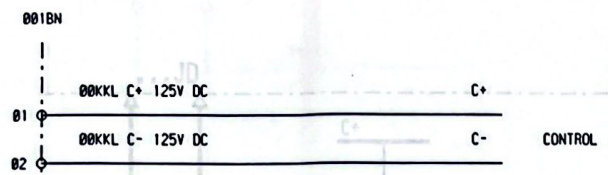
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ERE P

P2135A30113- A

04 07

BUIL2LE1303



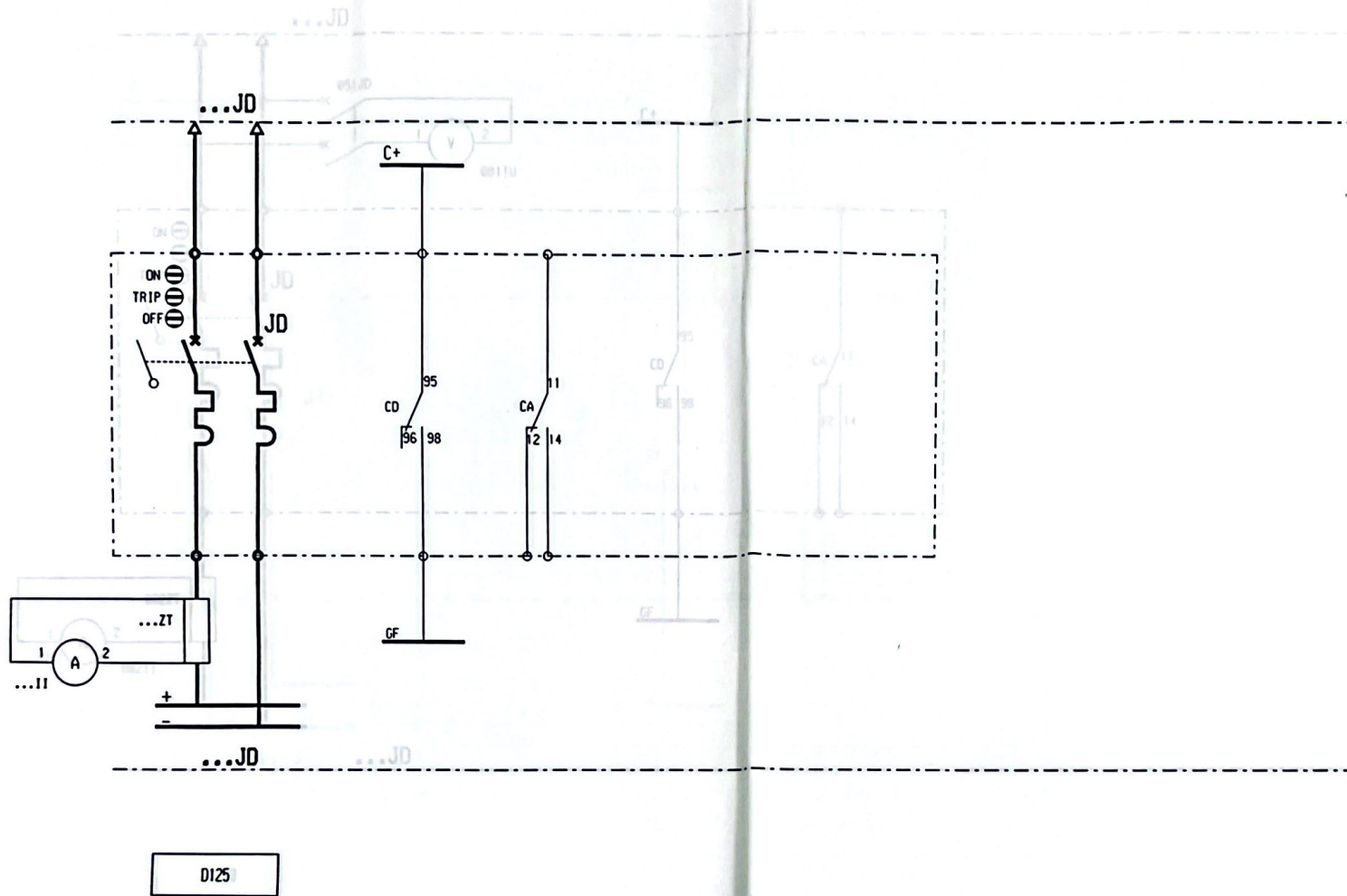
INCOMING CHARGER
POLARITIES DISTRIBUTION
2 POLES

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ERE P

P2135A30113- B

07 08

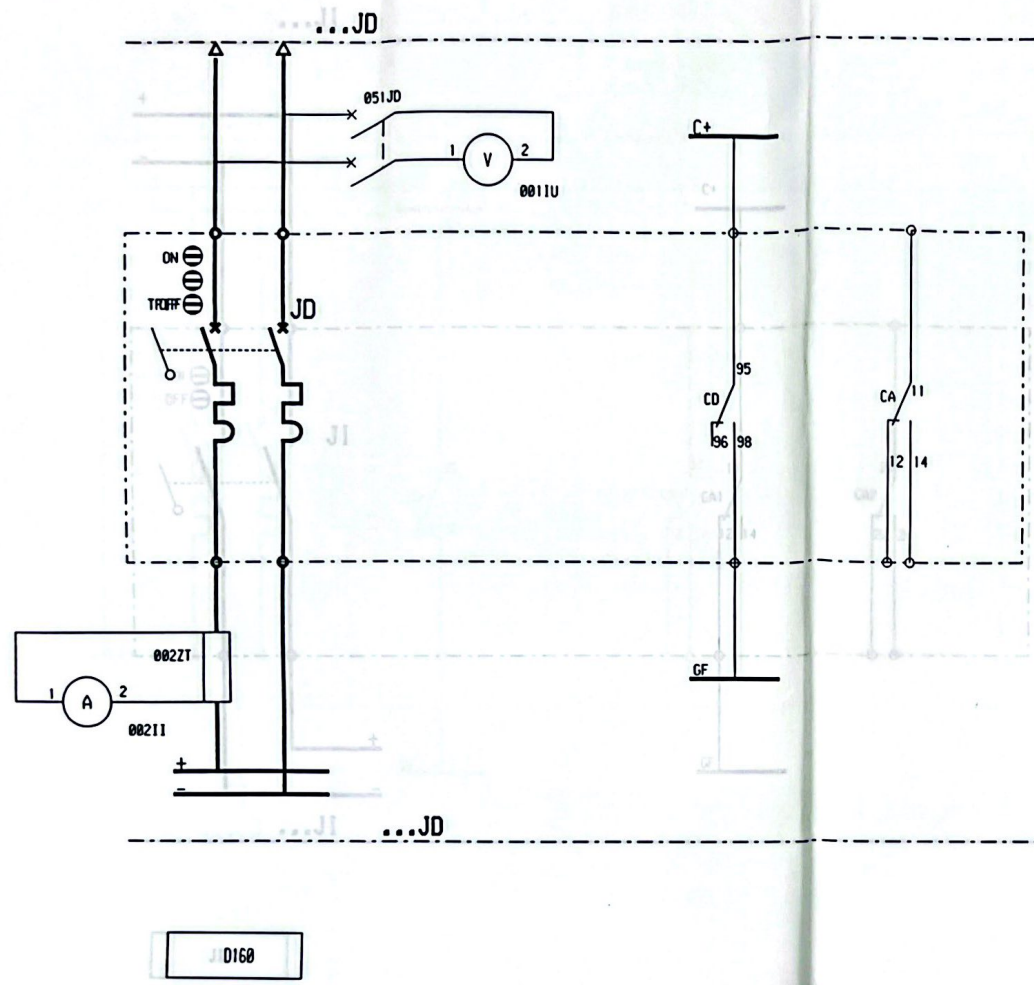


INCOMING CHARGER
MANUAL CIRCUIT BREAKER
2 POLES

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EQUIP
08 09
21

1 2 3 4 5 6 7 8

A
B
C
D
E
F



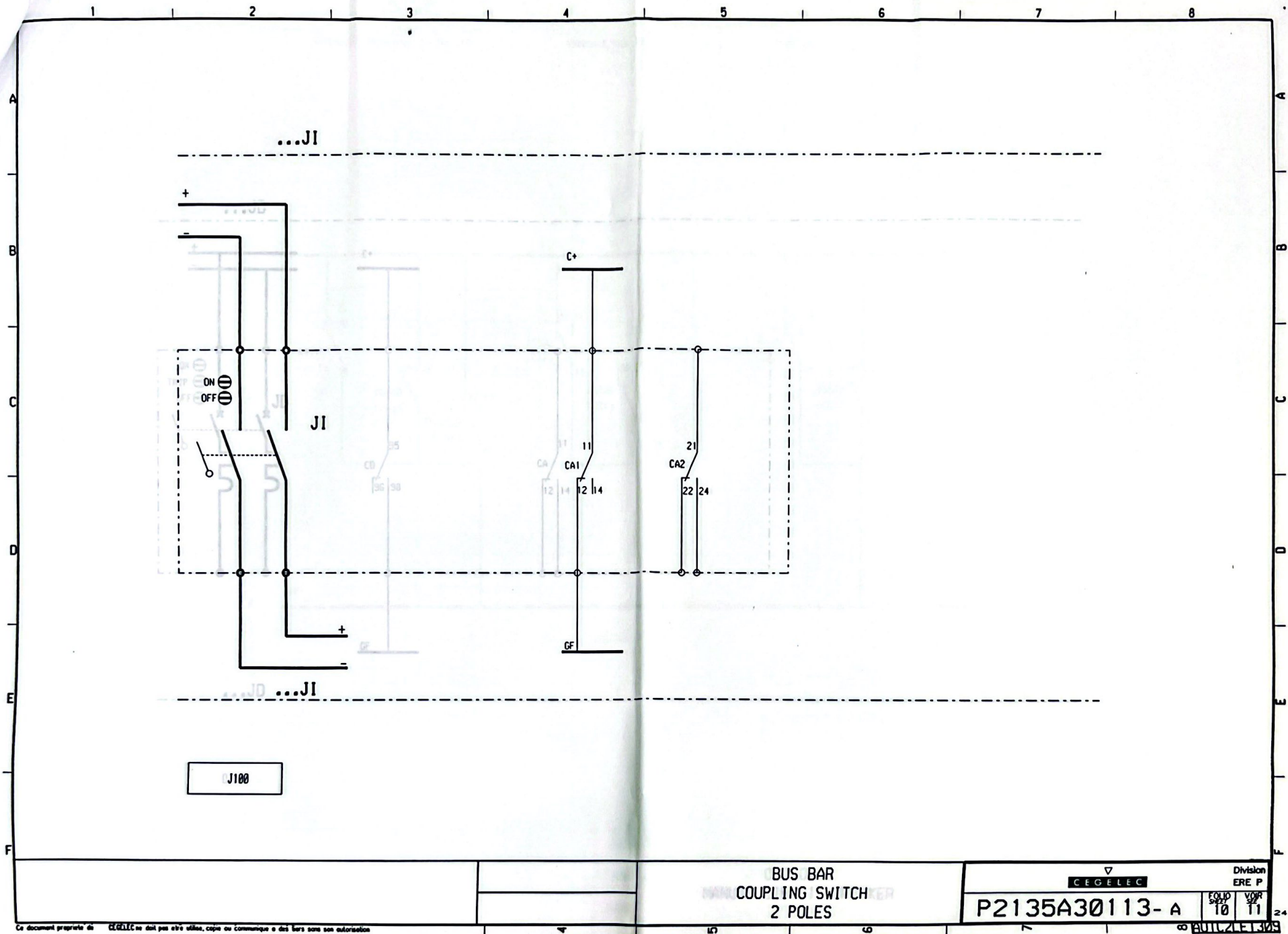
D160

INCOMING BATTERY
MANUAL CIRCUIT BREAKER
2 POLES

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P2135A30113-c
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23

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BUS BAR
COUPLING SWITCH
2 POLES

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P2135A30113-A

EQUID
10
VDR
11

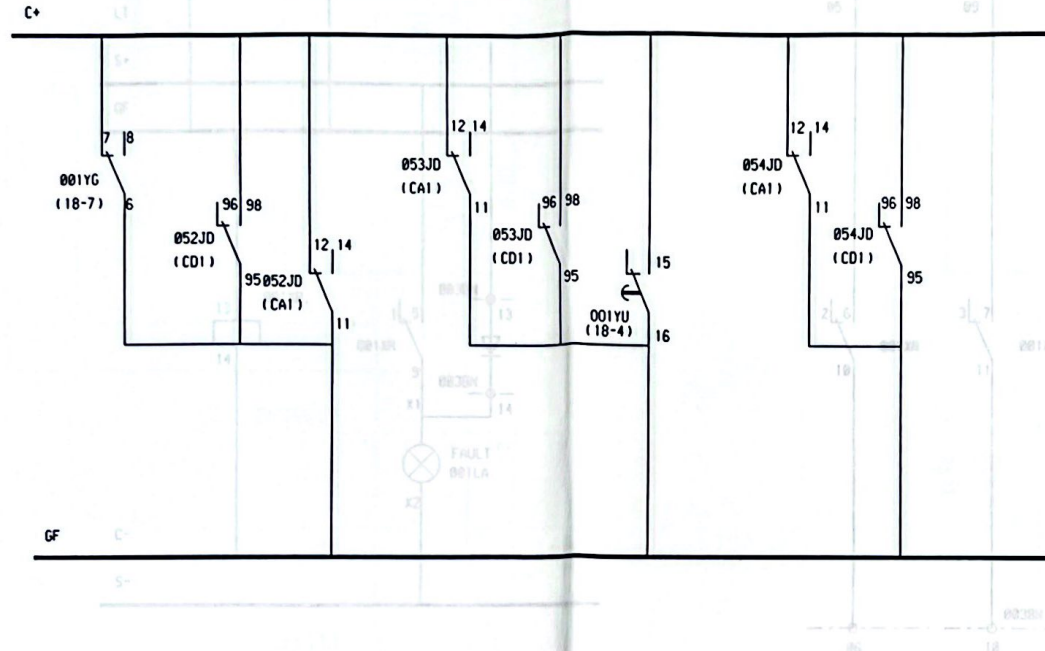
BUIL2LE1309

PERMANENT INSULATED CONTROLLER

UNDERVOLTAGE RELAY

POLARITY FAULT S+/-

SCHEMATIC FAULT MOTOR OVERLOAD (18-7)



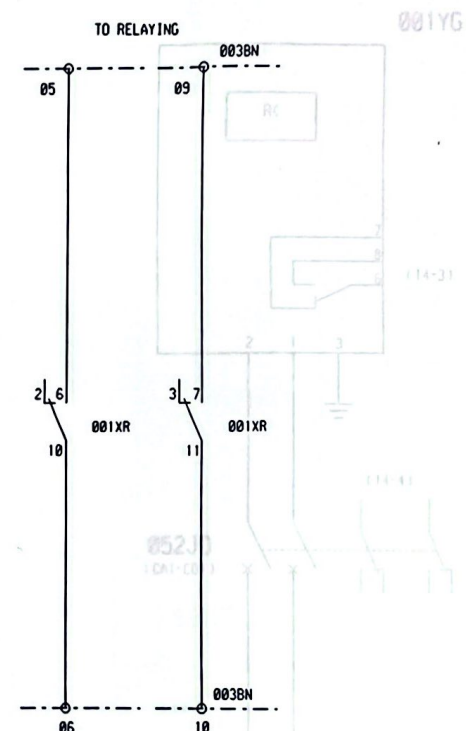
SWITCH POSITION
FUSE BLOWN - UNDERVOLTAGE
INSULATED CONTROLLER

CEGELEC

Division
ERE P

P2135A30113-B

14 15 12

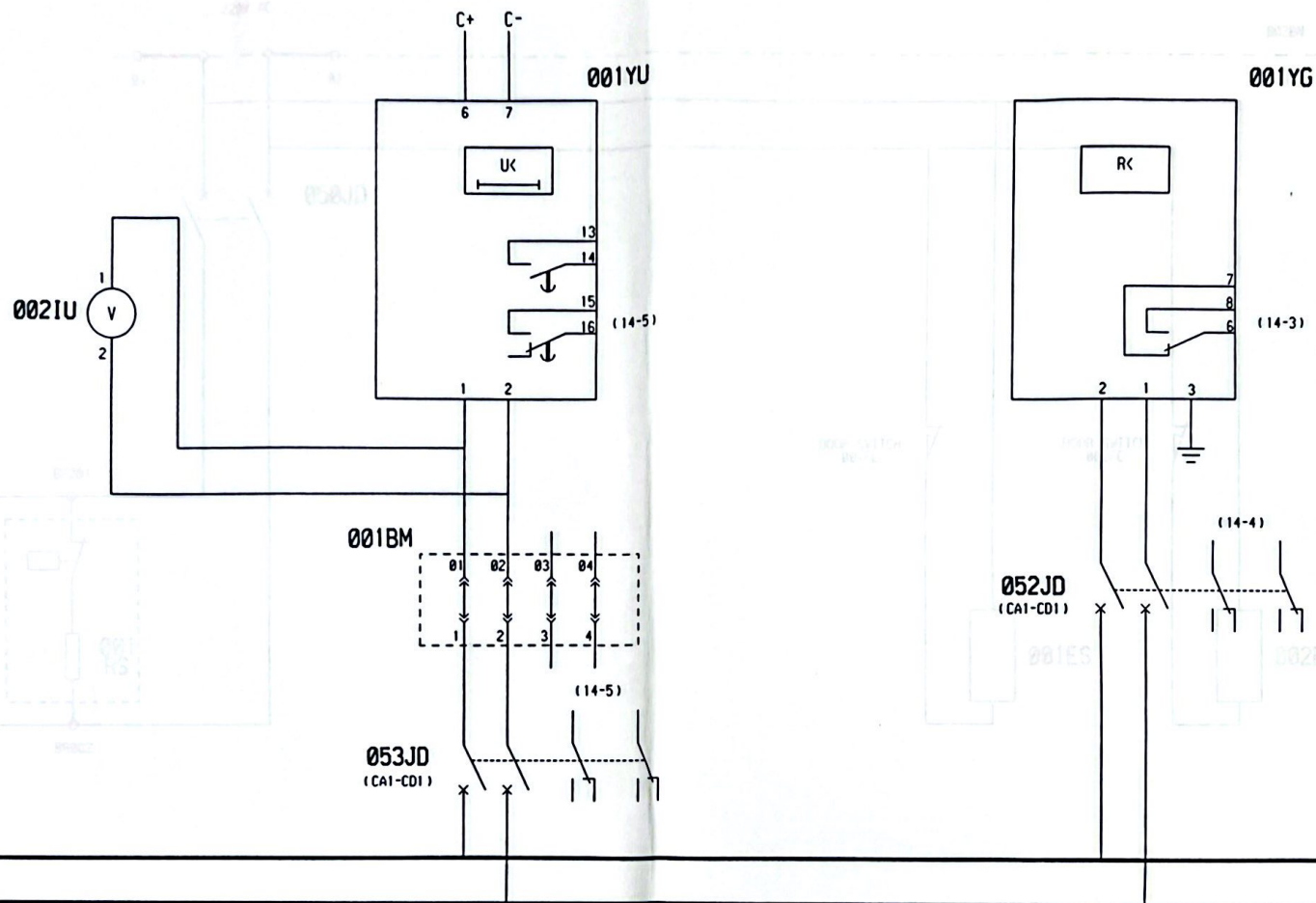


GROUPED FAULT SYSTEM

P2135A30113-B

Division
ERE P

FOLIO SHEET 15	VOR SEE 18
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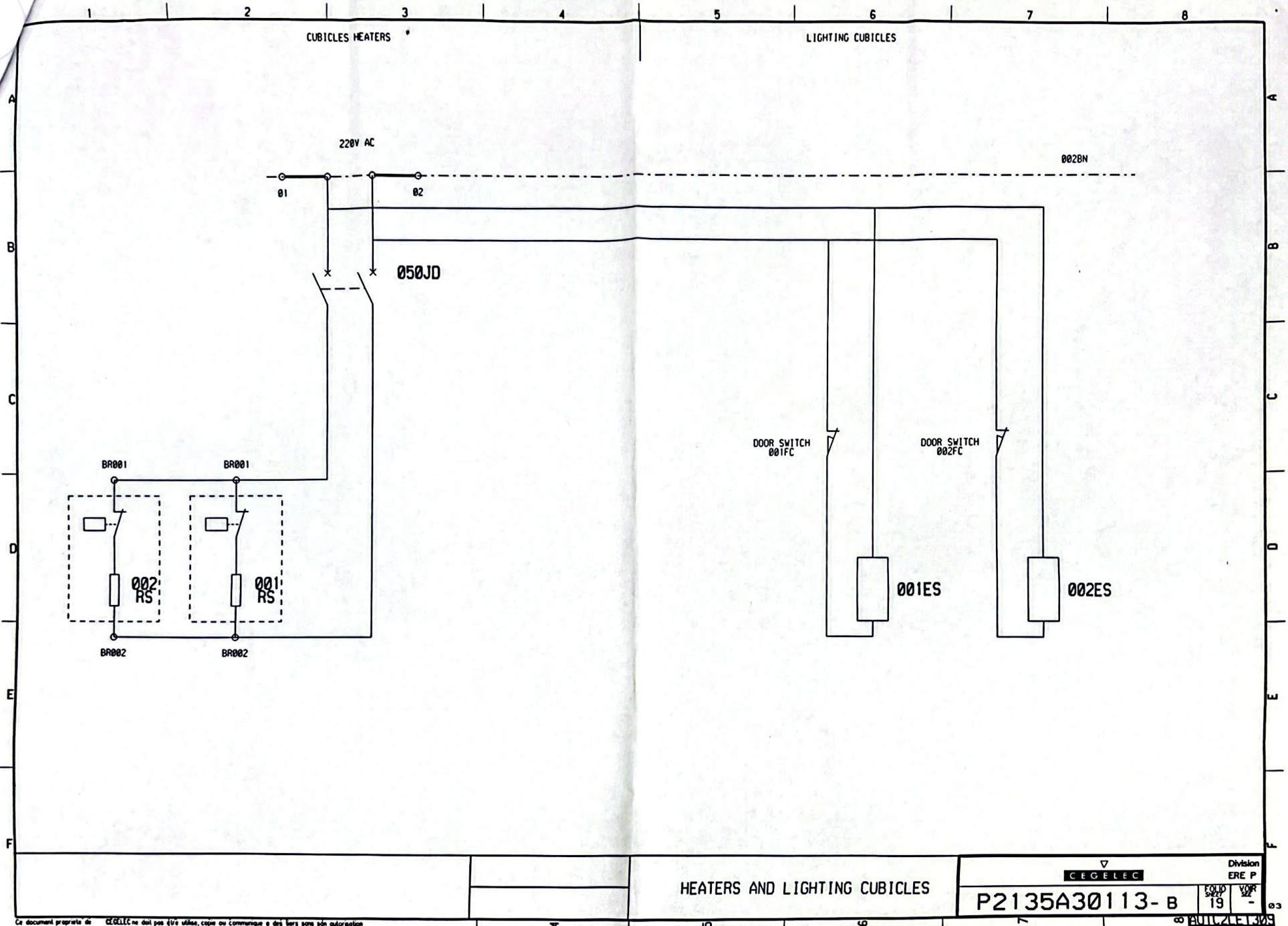
PROTECTIVE AND METERING
SYSTEM

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Division
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18 19

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